Bridging Critical AI Frameworks with Data Storage Practices: the AIAI Data Collective

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Data is central to the construction and maintenance of artificial intelligence. It is what models are trained on, and serves as the foundation for what and how they learn. As such, data is also fundamental to our understanding of the ethics and consequences of ever-emerging new models with continually developing capabilities. In recent years, scholars in the emerging field of Critical AI have made landmark arguments about the ethics, labor practices, and environmental impact of AI. These arguments primarily engage with the construction and significance of AI datasets– what they include, who they exclude, and their reflections of ongoing social contexts.[1] Yet little has been said about what to do with these datasets after they have been built. How does one store AI training data? What are the relative ethical implications of training data storage?

This project applies Critical AI frameworks to latent questions of AI data storage, and proposes a digital tool that will guide users through the decision-making process for ethical AI data management. Methodologically interdisciplinary, this project pulls from technical fields and library sciences, with a particular emphasis on humanities orientations. Specifically, we apply frameworks such as feminist theory, critical race theory, and other social, cultural, and economic frames to the issue of AI data storage. Our reflexive and analytic questioning practice encourages users to consider the labor practices, sustainability, privacy, and representational issues that underlie AI data storage.

This project was born out of the Atlanta Interdisciplinary AI Network (AIAI), a Mellon-funded initiative that applies humanistic methods to questions of data justice and AI ethics.[2] The AIAI Data Collective is one project within the Network, tasked with storing the data from other AIAI Network research projects.[3] While researching best practices for the Collective, we have identified scholarly gaps around AI data storage, such as the necessity of long term labor to ensure access and accountability. While public and scholarly discourse is actively attuned to the unfair conditions and devaluing of data labor globally, rarely has it devoted attention to the ways in which these problems exist not only in data creation and curation, but also in data storage.

Guided by the interdisciplinary and inter-institutional foundations of the AIAI Network, this project expands the bounds of Critical AI scholarship, making room for sociocultural frameworks alongside technical problems and solutions. The specificity of the data produced by AIAI researchers – which is primarily humanities-based and Atlanta-specific—also facilitated the development of this inquiry. While much of the work of Critical AI has been necessarily broad[4] – as AI training data itself is global and expansive – this project is able to approach AI data storage with context and specificity, which we argue is central to ethical AI data storage.

The AIAI Data Collective argues that ethical AI data storage relies on a reflective and reflexive decision-making process. To enact this argument, we have developed a digital tool which walks researchers through key questions about their data —on topics such as privacy and funding—and recommends specific practices, considerations, and questions based on their responses. Given the many, broad contexts of data production, and the unique ethical imperatives raised by each one, the objective of this tool is to identify primary concerns in the context of an individual research project, rather than proposing a set of best practices for data management in general. For instance, input regarding the user's institutional infrastructure, staffing, and ongoing funding will determine how active or static their data storage should be, and what governance models— like crowdsourcing, data commons, or individual management— might be available to them in the long or short-term.

Using projects like Timnit Gebru et al.'s "Datasheets for Datasets" as a model for our inquiry, our data storage tool generates practicable solutions to concerns that may at first appear theoretical, systemic, or simply difficult for researchers to prioritize. The tool addresses critical and ethical concerns such as the reification of bias, the devaluation of data work, and privacy and data ownership by synthesizing existing Critical AI scholarship with practical research in fields like Library and Information Sciences related to data storage and management. We argue that de-

cisions around issues like accessibility and governance will inevitably be shaped by institutional, financial, and other systemic constraints; and so, using the CARE and FAIR principles of data management to guide the tool's output, we provide guidance based on the user's unique circumstances, so they can make strategic and reflective decisions about how to put their principles and values into action. For example, drawing on Carroll et al.'s scholarship on operationalizing CARE principles in indigenous communities, we recommend that users tailor their decisions to the specific contexts from which their data was drawn, emphasizing criteria such as collective benefit, authority to control, and direct input (when necessary or possible) from a given community (Carroll et al 2021.).

This tool currently serves as the basis of the AIAI Data Collective's data storage procedures and practices. However, we intend to expand its scope to account for a wider range of AI datasets in the future. To apply the framework we put forth, this tool must capture the wide variety of structural, disciplinary, and proprietary limitations that accompany AI research– from institutional project sustainability to Big Data's funding and ownership of data research. This project will continue to iterate based on cross-disciplinary feedback as we enter phase 2 of this project and conduct focus groups with data scientists, CS scholars, community-oriented data justice leaders, and humanities scholars. Accounting for this wide array of fields and disciplines, we intend to address rising issues around ethics and data storage by expanding the reach of sociocultural frameworks to AI data practices and processes.

Notes

1. For example, see Joy Buolamwini and Timnit Gebru's "Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification" (2018) which found that classifier models have difficulty identifying women of color because their datasets primarily feature light-skinned people, Safiya Noble's *Algorithms of Oppression: How Search Engines Reinforce Racism* (2019), which argues that anti-Black racism is a data input as well as a model output for AI, and Nick Couldry and Ulysses A. Mejias's Data Grab: The New Colonialism of Big Tech and How to Fight Back (2024), which argues that current day data extraction is an extension of historical colonialism.
2. For more on the AIAI Network, read their mission statement at https://www.aiai.network/
3. Examples of these projects include "AI and Social Movements," (Project Lead: Lauren Klein, Emory University, Department of English) which uses #StopCopCity and #BlackLivesMatter as case studies to explore the uses and limits of AI for the study of social movements, the "Evictions Counter Data Project," (Project Lead: Anh-Ton Tran, Georgia Tech, School of Interactive Computing) which uses design, theory, and ethnography to explore how data is created and circulated through the eviction process in Atlanta, and "Public Interest AI" (Project Lead: Brandeis Marshall, DataEdX) which examines and critiques the interlocking historical, social, political, and technical aspects of the city's data quality management in relation to the Atlanta Beltline Project.
4. For instance, the seminal 2021 article "On the Dangers of Stochastic Parrots" responds to multiple AI models (BERT, GPT-2/3, etc.) in order to critique industry-wide trends and concerns. As smaller models have begun to proliferate in more recent years—not to mention proprietary and scholarly use of secondary datasets—specificity of analysis has become necessary in addition to the larger-scale criticisms contributed by Bender, Gebru, et al.

Bibliography

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